Assignment 5: Data Visualization

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OVERVIEW

This exercise accompanies the lessons in Environmental Data Analytics on Data Visualization

Directions

1. Rename this file `<FirstLast>_A05_DataVisualization.Rmd` (replacing `<FirstLast>` with your first and last name).
2. Change "Student Name" on line 3 (above) with your name.
3. Work through the steps, **creating code and output** that fulfill each instruction.
4. Be sure your code is tidy; use line breaks to ensure your code fits in the knitted output.
5. Be sure to **answer the questions** in this assignment document.
6. When you have completed the assignment, **Knit** the text and code into a single PDF file.

Set up your session

1. Set up your session. Load the tidyverse, here & cowplot packages, and verify your home directory. Read in the NTL-LTER processed data files for nutrients and chemistry/physics for Peter and Paul Lakes (use the tidy `NTL-LTER_Lake_Chemistry_Nutrients_PeterPaul_Processed.csv` version in the Processed_KEY folder) and the processed data file for the Niwot Ridge litter dataset (use the `NEON_NIWO_Litter_mass_trap_Processed.csv` version, again from the Processed_KEY folder).

2. Make sure R is reading dates as date format; if not change the format to date.

```
#1
library(tidyverse); library(here); library(cowplot); library(ggthemes)
```

```
## -- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
## v dplyr     1.1.4     v readr     2.1.5
## v forcats   1.0.0     v stringr   1.5.1
## v ggplot2   3.5.1     v tibble    3.2.1
## v lubridate 1.9.3     v tidyr     1.3.1
## v purrr     1.0.2
## -- Conflicts ------------------------------------------ tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()    masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
## here() starts at /home/guest/project/EDE_Fall2025
##
##
## Attaching package: 'cowplot'
##
##
## The following object is masked from 'package:lubridate':
##
##     stamp
##
##
##
## Attaching package: 'ggthemes'
##
##
## The following object is masked from 'package:cowplot':
##
##     theme_map
```

```
here()
```

```
## [1] "/home/guest/project/EDE_Fall2025"
```

```
PeterPauldf <- read.csv(here('Data',
                             'Processed_KEY',
                             "NTL-LTER_Lake_Chemistry_Nutrients_PeterPaul_Processed.csv"),
                        stringsAsFactors = T)

NeonNiwotdf <- read.csv(here('Data',
                             'Processed_KEY',
                             "NEON_NIWO_Litter_mass_trap_Processed.csv"),
                        stringsAsFactors = T)

#2
PeterPauldf$sampledate <- ymd(PeterPauldf$sampledate)
NeonNiwotdf$collectDate <- ymd(NeonNiwotdf$collectDate)
```

Define your theme

3. Build a theme and set it as your default theme. Customize the look of at least two of the following:

- Plot background
- Plot title
- Axis labels
- Axis ticks/gridlines
- Legend

```
#3
my_theme <- theme_base()  +
  theme(
    #line =              element_line(),
```

```
    #rect =              element_rect(),
    #text =              element_text(),

    # Modified inheritance structure of text element
    plot.title =       element_text(family = "mono",
                                    face = "bold"),
    axis.title.x =     element_text(family="mono",
                                     face = "italic"),
    axis.title.y =     element_text(family = "mono",
                                     face = "italic"),
    # Modified inheritance structure of line element
    axis.ticks =       element_line(lineend = "butt"),
    panel.grid.major = element_line(color = "lightgrey",
                                    lineend = "butt"),
    #panel.grid.minor =  element_blank(),

    # Modified inheritance structure of rect element
    plot.background =  element_rect(fill="azure"),
    panel.background = element_rect(fill = "azure3"),
    #legend.key =        element_rect(),

    # Modifiying legend.position
    legend.position = 'top',

    #complete = TRUE
    )
```

Create graphs

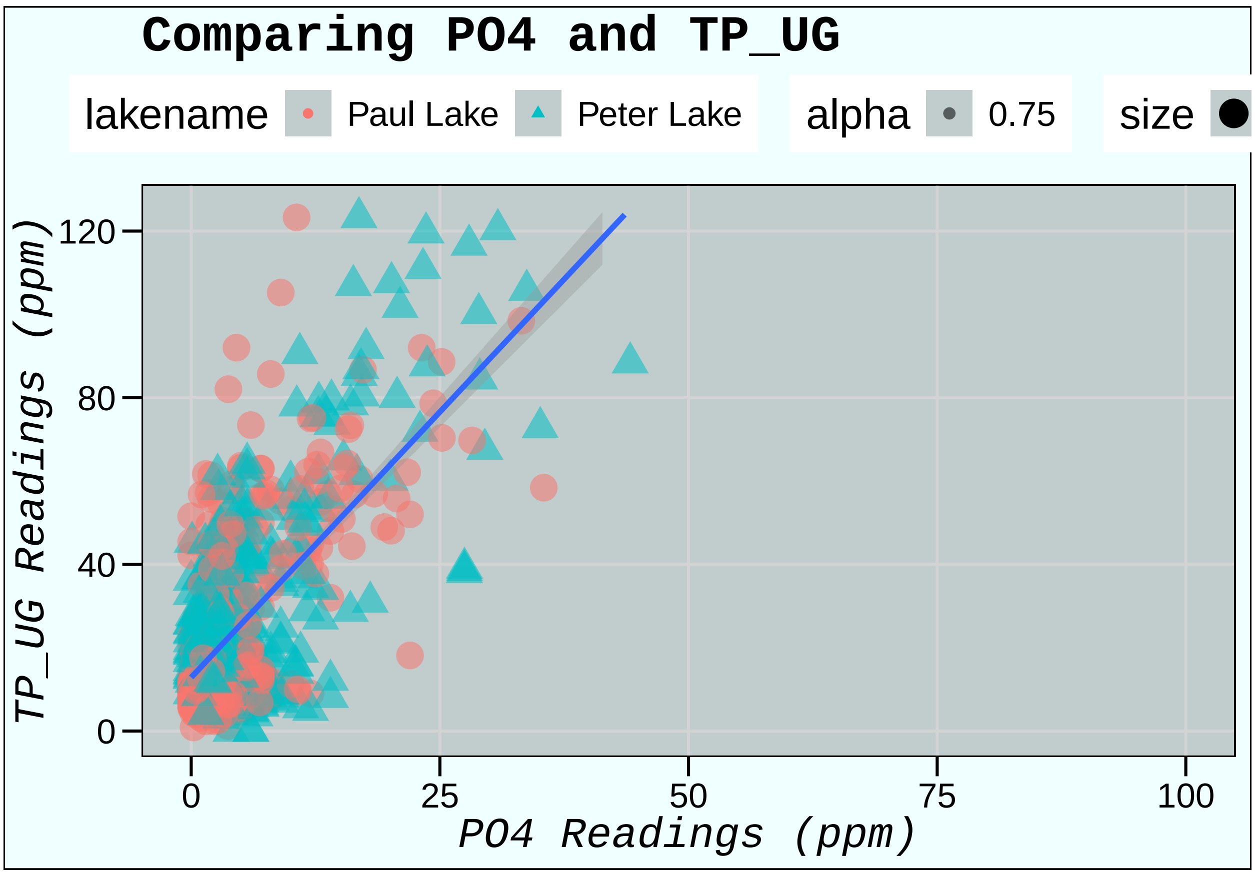
For numbers 4-7, create ggplot graphs and adjust aesthetics to follow best practices for data visualization. Ensure your theme, color palettes, axes, and additional aesthetics are edited accordingly.

4. [NTL-LTER] Plot total phosphorus (`tp_ug`) by phosphate (`po4`), with separate aesthetics for Peter and Paul lakes. Add line(s) of best fit using the `lm` method. Adjust your axes to hide extreme values (hint: change the limits using `xlim()` and/or `ylim()`).

```
#4
task4 <- ggplot(PeterPauldf,
                aes(x=po4,
                 y=tp_ug))+
  geom_point(aes(color = lakename,
                 shape=lakename,
                 size = 3,
                 alpha = 0.75))+
  geom_smooth(method="lm")+
  my_theme+
  xlim(0, 100) +
  ylim(0, 125)+
  labs(title = "Comparing PO4 and TP_UG")+
  xlab("PO4 Readings (ppm)")+
  ylab("TP_UG Readings (ppm)")
```

```
task4
```

```
## `geom_smooth()` using formula = 'y ~ x'

## Warning: Removed 21952 rows containing non-finite outside the scale range
## (`stat_smooth()`).

## Warning: Removed 21952 rows containing missing values or values outside the scale range
## (`geom_point()`).

## Warning: Removed 1 row containing missing values or values outside the scale range
## (`geom_smooth()`).
```



5. [NTL-LTER] Make three separate boxplots of (a) temperature, (b) TP, and (c) TN, with month as the x axis and lake as a color aesthetic. Then, create a cowplot that combines the three graphs. Make sure that only one legend is present and that graph axes are aligned.

Tips: * Recall the discussion on factors in the lab section as it may be helpful here. * Setting an axis title in your theme to `element_blank()` removes the axis title (useful when multiple, aligned plots use the same axis values) * Setting a legend's position to "none" will remove the legend from a plot. * Individual plots can have different sizes when combined using `cowplot`.

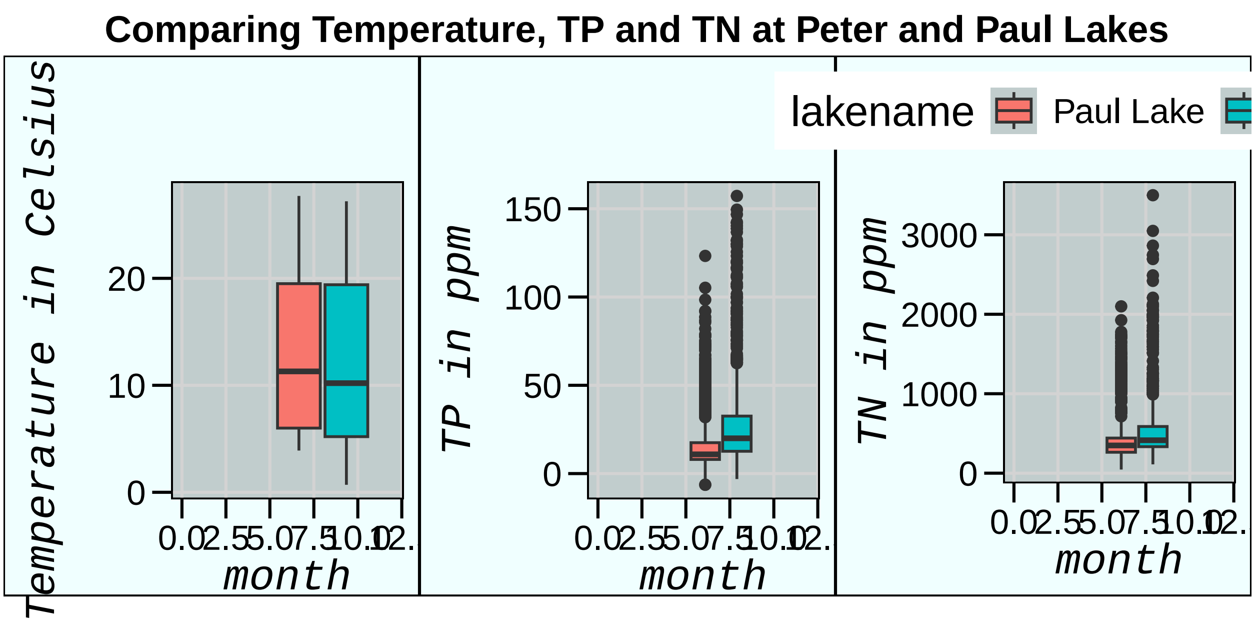
```r
#5

task5a <- ggplot(PeterPauldf)+
  geom_boxplot(aes(x=month, y=temperature_C, fill = lakename))+
  xlim(0,12)+
  my_theme+
  theme(legend.position = "none",
        plot.margin = margin(0,0,0,10))+
  ylab("Temperature in Celsius")
task5b <- ggplot(PeterPauldf)+
  geom_boxplot(aes(x=month, y=tp_ug, fill = lakename))+
  xlim(0,12)+
  my_theme+
  theme(legend.position = "none",
          plot.margin = margin(0,0,0,10))+
  ylab("TP in ppm")
task5c <- ggplot(PeterPauldf)+
  geom_boxplot(aes(x=month, y=tn_ug, fill = lakename))+
  theme(legend.position = "none",
          plot.margin = margin(0,0,0,10))+
  my_theme+
  xlim(0,12)+
  ylab("TN in ppm")

rowof3 <- plot_grid(task5a,
                    task5b,
                    task5c,
                    nrow = 1,
                    align = "hv",
                    axis="t",
                    rel_widths = c(.75,.75,.75))
```

```
## Warning: Removed 3566 rows containing non-finite outside the scale range
## (‘stat_boxplot()‘).

## Warning: Removed 20729 rows containing non-finite outside the scale range
## (‘stat_boxplot()‘).

## Warning: Removed 21583 rows containing non-finite outside the scale range
## (‘stat_boxplot()‘).
```

```r
# code below borrowed from cowplot website.

legend <- get_legend(task5c
                     +
                       theme(legend.box.margin = margin(0, 0, 0, 12)))
```

```
## Warning: Removed 21583 rows containing non-finite outside the scale range
## (‘stat_boxplot()‘).

## Warning in get_plot_component(plot, "guide-box"): Multiple components found;
## returning the first one. To return all, use ‘return_all = TRUE‘.
```

```r
Task5Title <- ggdraw()+
  draw_label(
    "Comparing Temperature, TP and TN at Peter and Paul Lakes",
    fontface = "bold",
  ) +
  theme(
    plot.margin=margin(0,0,0,7)
  )
plot_grid(Task5Title, rowof3, legend, ncol=1,
          rel_heights = c(0.1,1, .5))
```



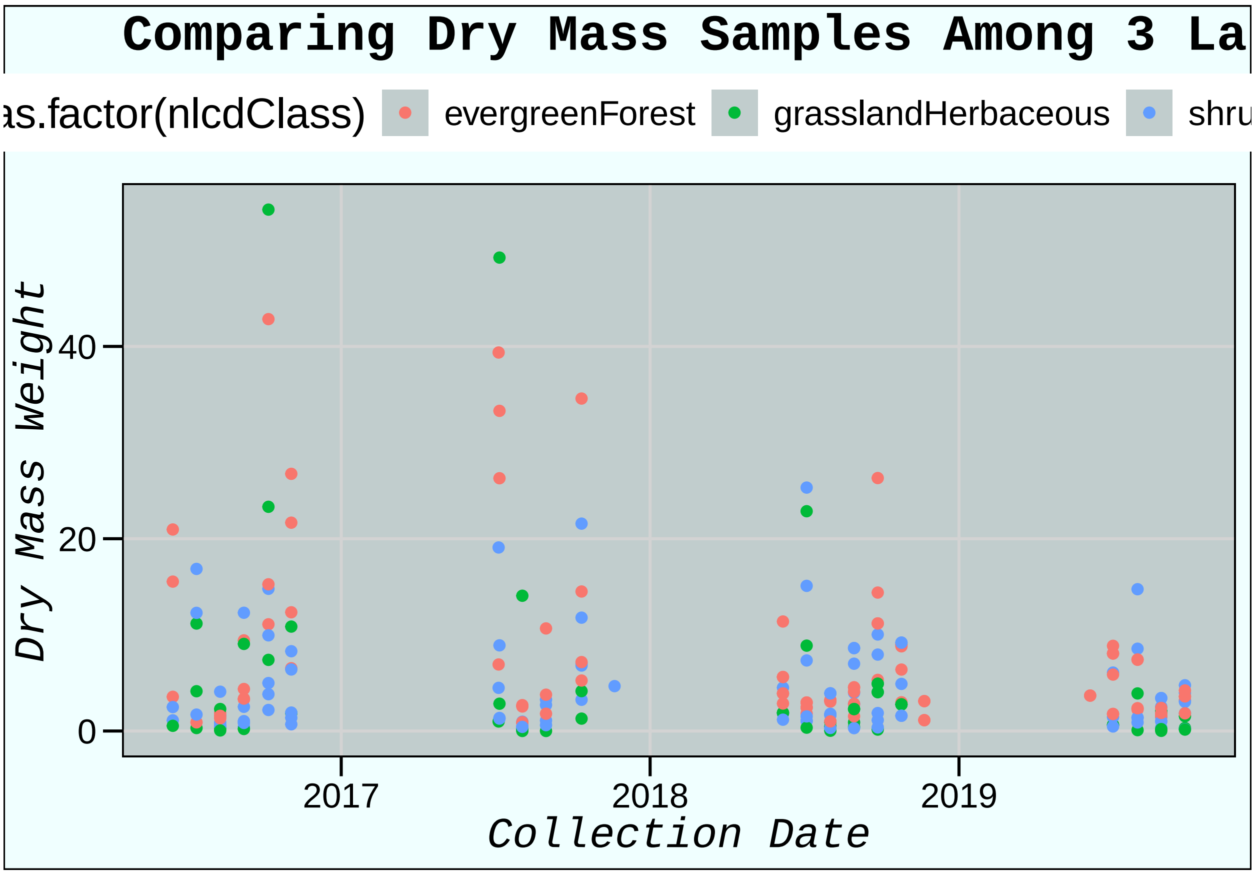
Question: What do you observe about the variables of interest over seasons and between lakes?

> Answer: Interestingly, Peter Lake's data has a longer skew towards cold temperatures, and much longer tails of extreme values in TP and TN.

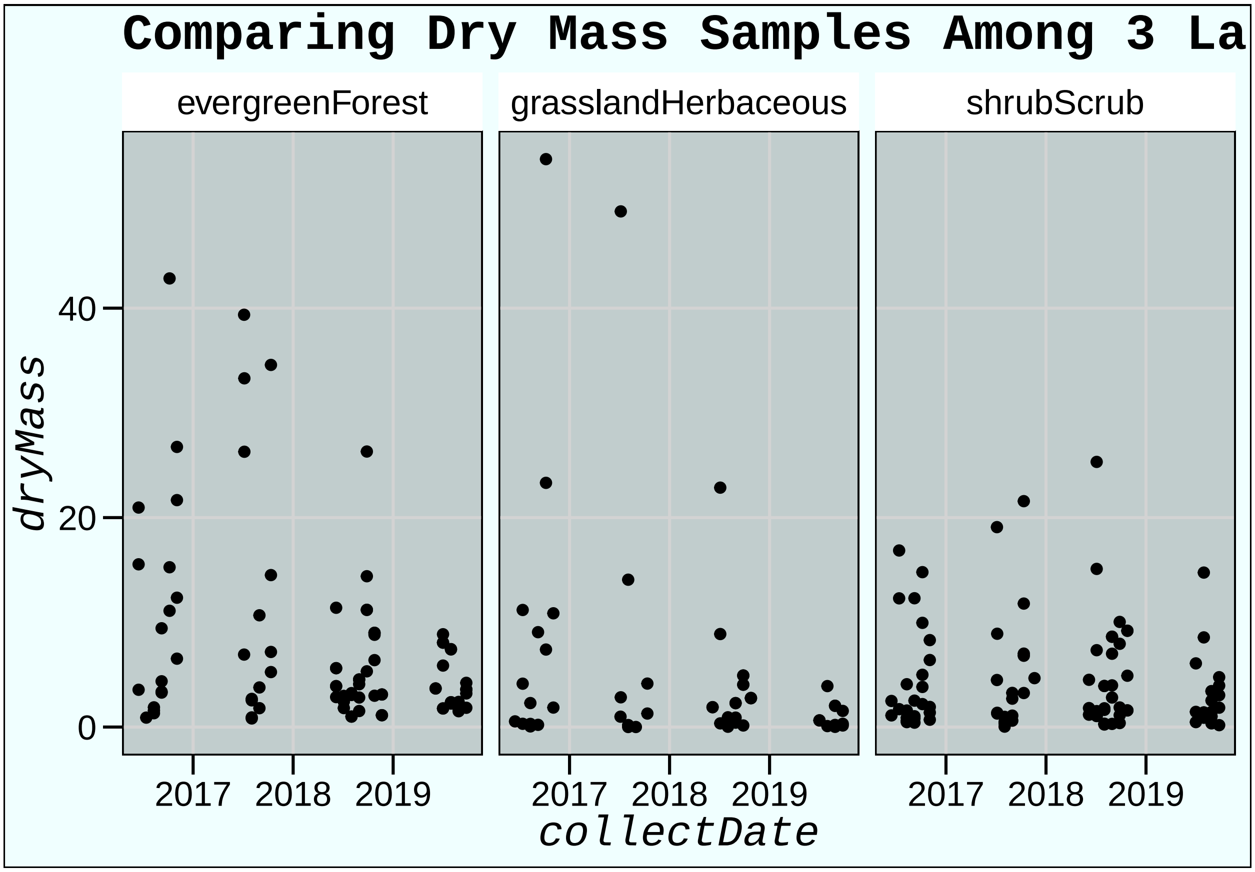
6. [Niwot Ridge] Plot a subset of the litter dataset by displaying only the "Needles" functional group. Plot the dry mass of needle litter by date and separate by NLCD class with a color aesthetic. (no need to adjust the name of each land use)
7. [Niwot Ridge] Now, plot the same plot but with NLCD classes separated into three facets rather than separated by color.

```r
#6
task6 <- ggplot(subset(NeonNiwotdf, functionalGroup == "Needles"))+
```

```
  geom_point(aes(x=collectDate, y=dryMass, color = as.factor(nlcdClass)))+
  xlab("Collection Date")+
  ylab("Dry Mass Weight")+
  theme(legend.title= "Land Cover Types")+
  labs(title = "Comparing Dry Mass Samples Among 3 Land Covers",
       fill = "Land Cover Types")+
  my_theme


task6
```




```
#7


task7 <- ggplot(subset(NeonNiwotdf, functionalGroup == "Needles"))+
  geom_point(aes(x=collectDate, y=dryMass))+
  theme(legend.title= "Land Cover Types")+
  labs(title = "Comparing Dry Mass Samples Among 3 Land Covers",
       fill = "Land Cover Types")+
  facet_wrap(vars(nlcdClass))+
  my_theme


task7
```



Question: Which of these plots (6 vs. 7) do you think is more effective, and why?

Answer: despite attempting to differentiate the different land covers by color in Task 6, I believe Task 7 is a much clearer depiction of how the data differ because there is less overlap and more ability to assess each data series separately.